

ASSIGNMENT-7.1

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Batch:36

Task Description #1 (Syntax Errors – Missing Parentheses in Print Statement)

Task: Provide a Python snippet with a missing parenthesis in a print statement (e.g., `print "Hello"`). Use AI to detect and fix the syntax error.

Bug: Missing parentheses in print statement

```
def greet():
```

```
print "Hello, AI Debugging Lab!"
```

```
greet()
```

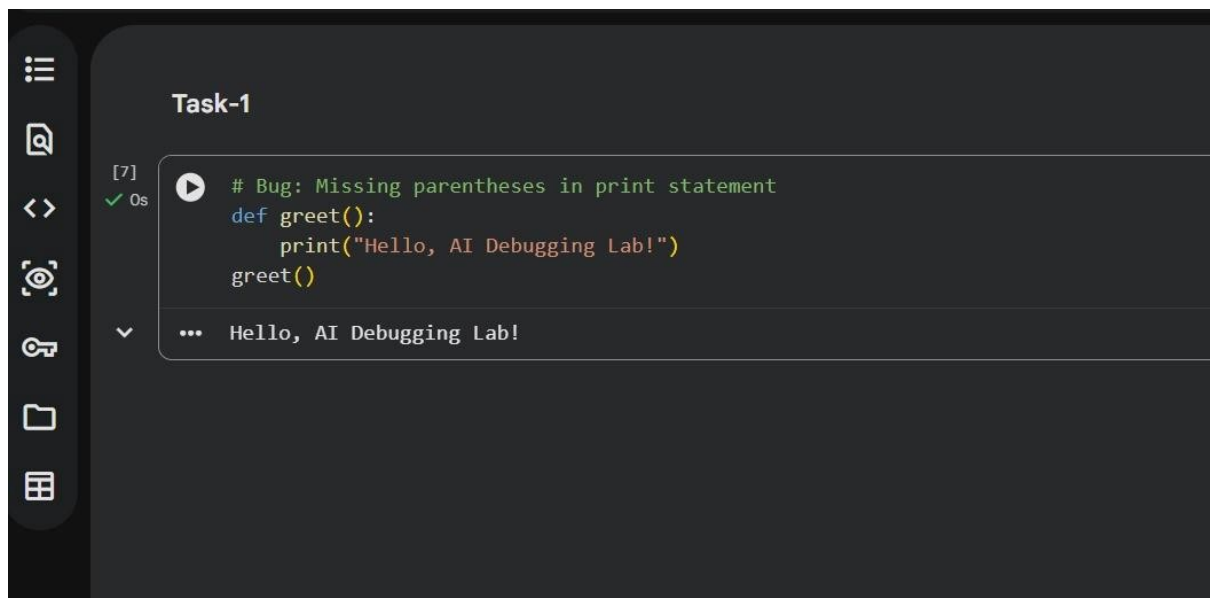
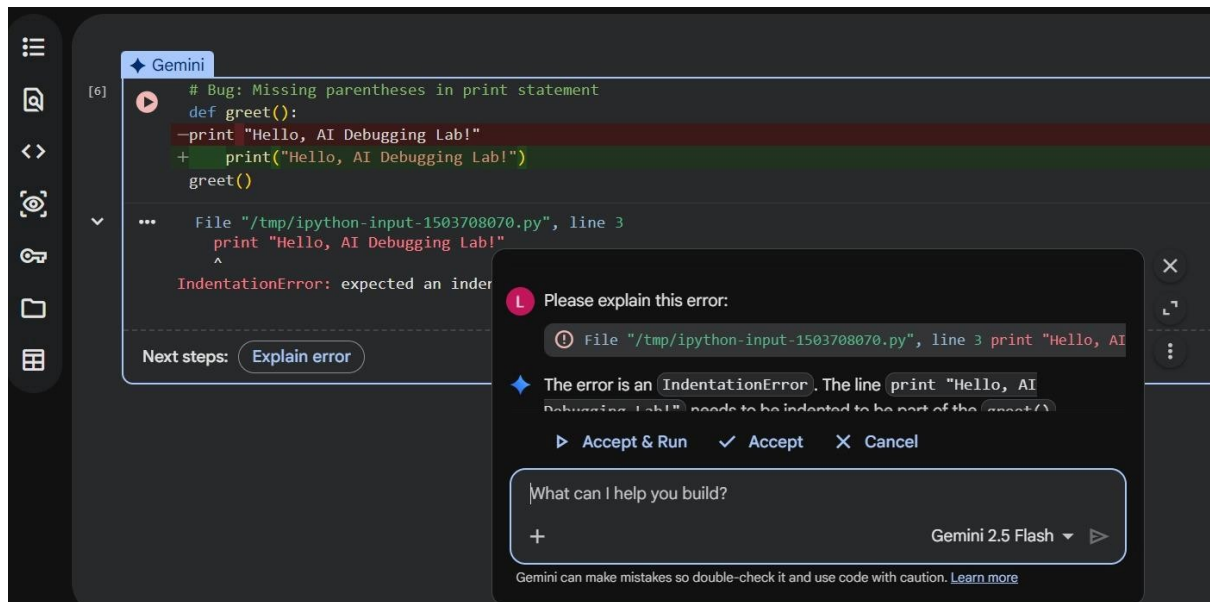
Requirements:

- Run the given code to observe the error.
- Apply AI suggestions to correct the syntax.
- Use at least 3 assert test cases to confirm the corrected code works.

Expected Output #1:

- Corrected code with proper syntax and AI explanation.

Output:



Task Description #2 (Incorrect condition in an If Statement)

Task: Supply a function where an if-condition mistakenly uses `=` instead of `==`. Let AI identify and fix the issue. # Bug: Using assignment (`=`) instead of comparison (`==`)

```
def check_number(n):
    if n = 10:
        return "Ten"
    else:
        return "Not Ten"
```

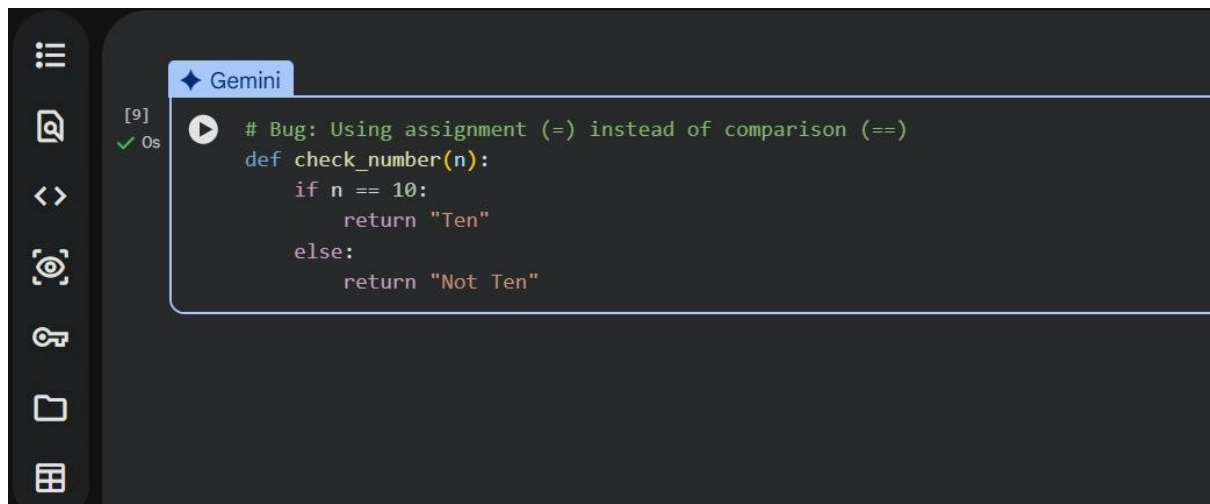
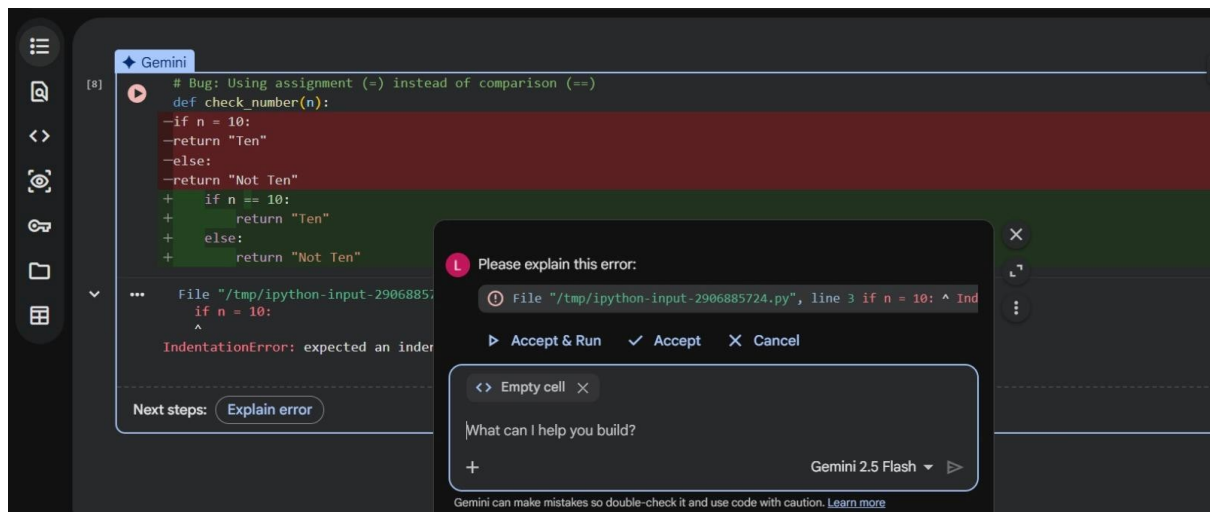
Requirements:

- Ask AI to explain why this causes a bug.
- Correct the code and verify with 3 assert test cases.

Expected Output #2:

- Corrected code using == with explanation and successful test execution.

Output:



Task Description #3 (Runtime Error – File Not Found)

Task: Provide code that attempts to open a non-existent file and crashes. Use AI to apply safe error handling. # Bug: Program crashes if file is missing

```
def read_file(filename):
    with open(filename, 'r') as f:
```

```
return f.read()

print(read_file("nonexistent.txt"))
```

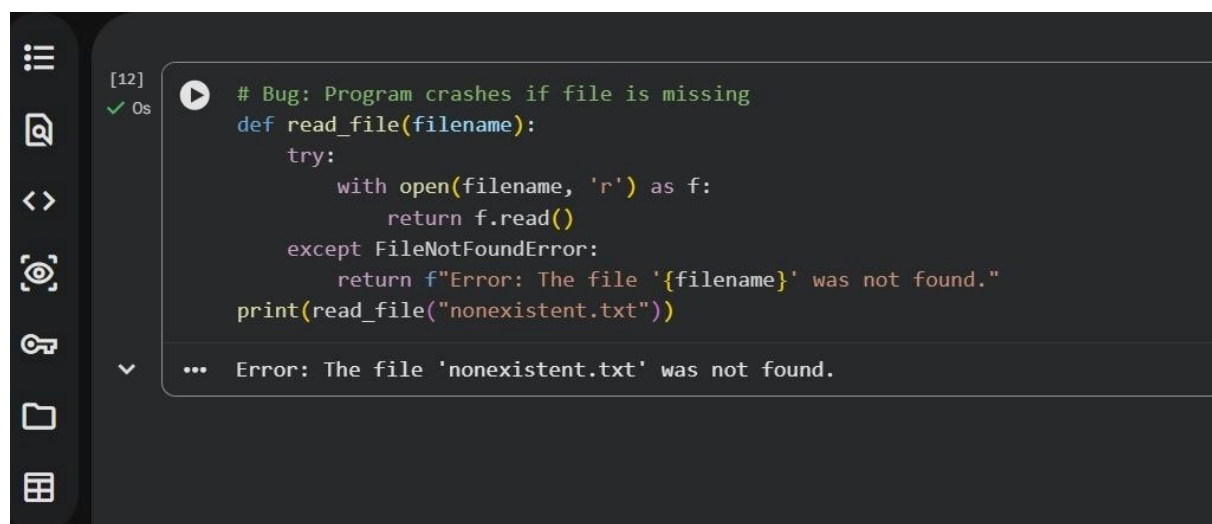
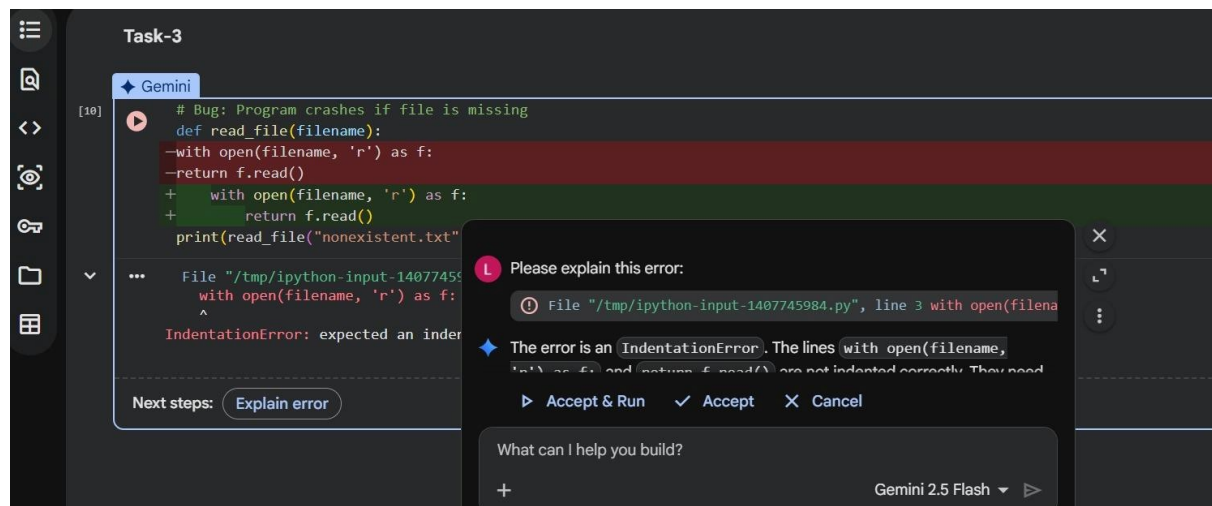
Requirements:

- Implement a try-except block suggested by AI.
- Add a user-friendly error message.
- Test with at least 3 scenarios: file exists, file missing, invalid path.

Expected Output #3:

- Safe file handling with exception management.

Output:



Task Description #4 (Calling a Non-Existent Method) Task:

Give a class where a non-existent method is called (e.g., `obj.undefined_method()`). Use AI to debug and fix.

Bug: Calling an undefined

method class Car: def start(self):

return "Car started" my_car = Car()

print(my_car.drive()) # drive() is not defined

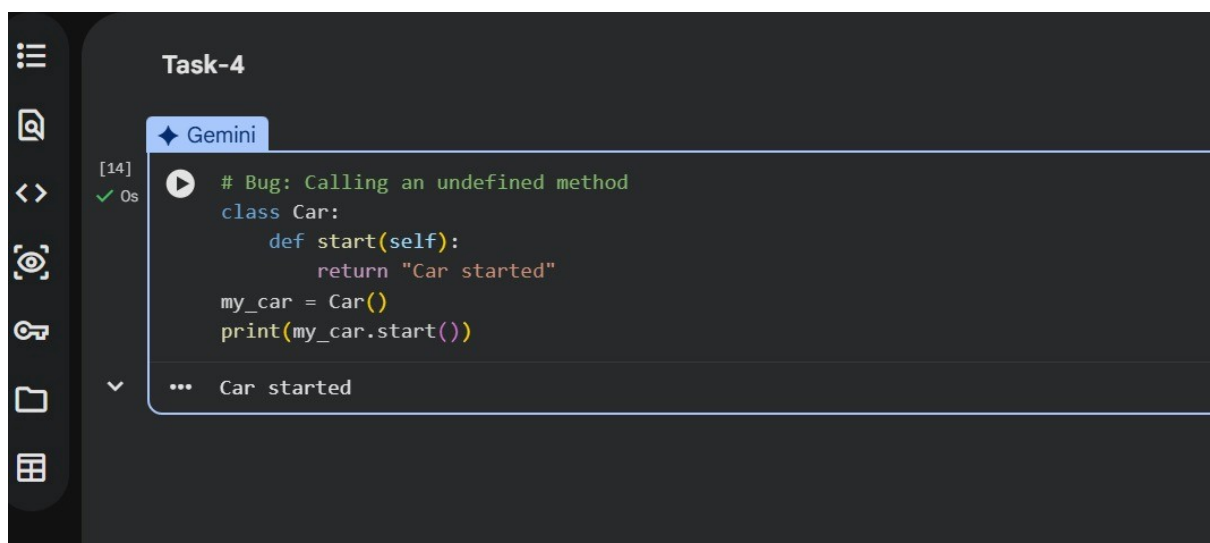
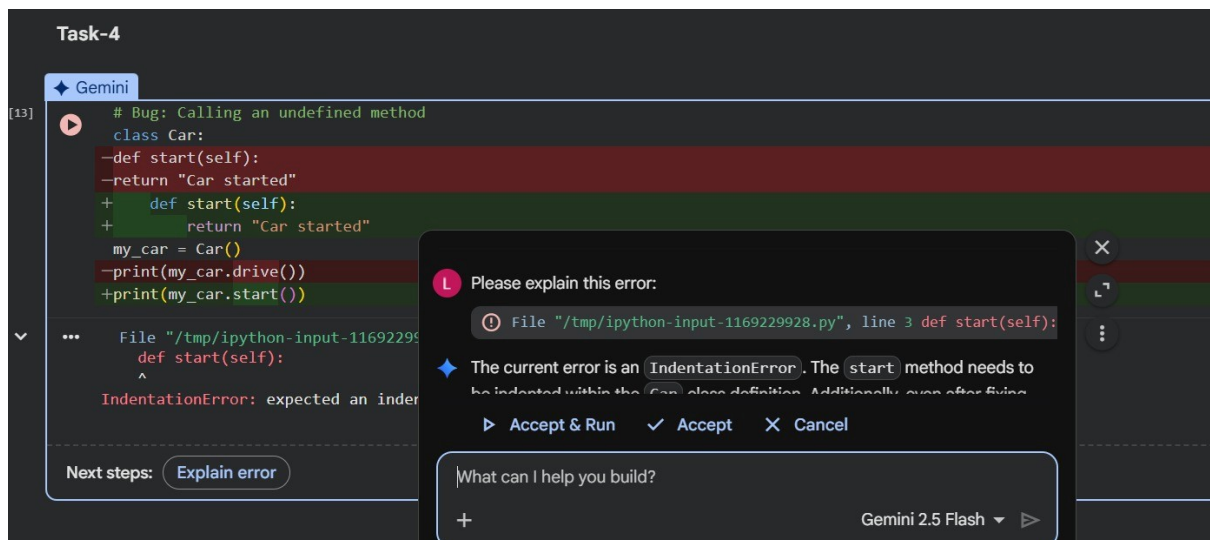
Requirements:

- Students must analyze whether to define the missing method or correct the method call.
- Use 3 assert tests to confirm the corrected class works.

Expected Output #4:

- Corrected class with clear AI explanation.

Output:



Task Description #5 (TypeError – Mixing Strings and Integers in Addition)

Task: Provide code that adds an integer and string ("5" + 2) causing a TypeError. Use AI to resolve the bug.

Bug: TypeError due to mixing string and integer

```
def add_five(value):
    return value + 5

print(add_five("10"))
```

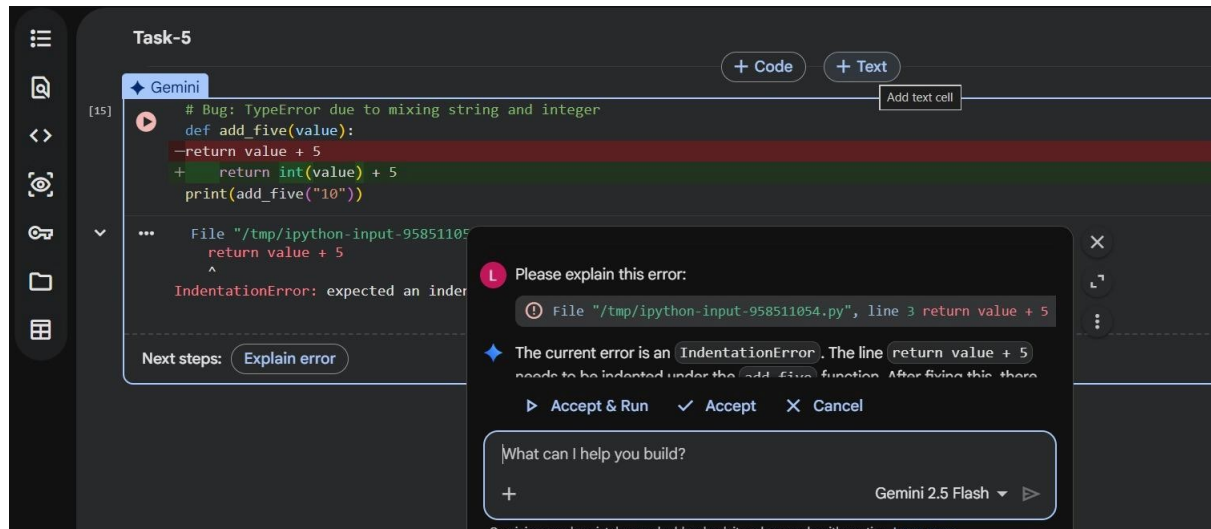
Requirements:

- Ask AI for two solutions: type casting and string concatenation.
- Validate with 3 assert test cases.

Expected Output #5:

- Corrected code that runs successfully for multiple inputs.

Output:

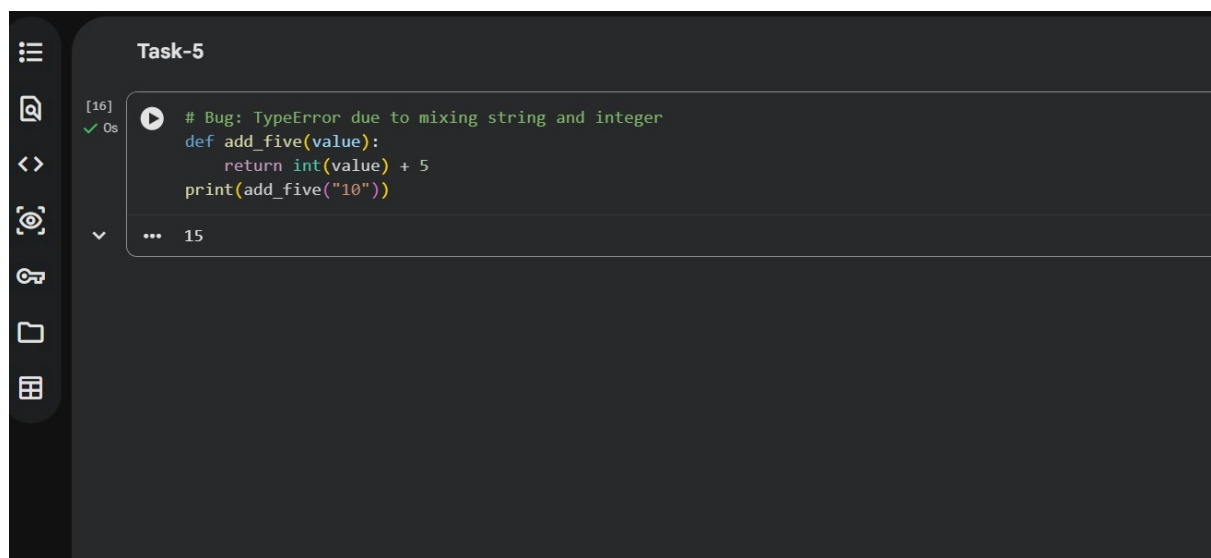


The screenshot shows a code editor window titled "Task-5". The code defines a function `add_five` and calls it with the string "10". A red squiggly line under the `return value + 5` line indicates an error. The error message in the console is `IndentationError: expected an indented block`. An AI assistant (Gemini) has provided an explanation: "Please explain this error: File '/tmp/ipython-input-958511054.py', line 3 `return value + 5`". The assistant explains that the line needs to be indented under the `add_five` function. The code is as follows:

```
[15] # Bug: TypeError due to mixing string and integer
def add_five(value):
- return value + 5
+ return int(value) + 5
print(add_five("10"))
```

Next steps: [Explain error](#)

What can I help you build? Gemini 2.5 Flash



The screenshot shows the same code editor window after the code has been corrected and executed. The `return value + 5` line is now correctly indented to `return int(value) + 5`. The output in the console is `15`. The code is as follows:

```
[16] # Bug: TypeError due to mixing string and integer
✓ 0s def add_five(value):
      return int(value) + 5
print(add_five("10"))

... 15
```